Software Engineering

Introduction – 24 February 2025

Content

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Team

- Course:
  - Adrian Iftene adiftene@info.uaic.ro
  - Alex Moruz mmoruz@info.uaic.ro
- Labs:
  - Adrian Iftene adiftene@info.uaic.ro
  - Cristian Simionescu cristian@nexusmedia.ro

Course content

- Software engineering
- Developing models
- Requirements engineering
- UML diagrams
- Design patterns
- Testing and debugging
- Maintenance
- Software metrics
- Project management
- Intellectual property and rights

Courses (2)

- **Why attend courses?**

- To better understand theoretical aspects + case studies
- 40 of 240 points for the exam come from questions which are discussed, without appearing on slides
- Bonuses… Presentations…

Laboratories

- UML diagrams, Design Patterns, Unit testing
- Implementations in Java, C++, C#, OOP (coding style)
- Topics proposed by the teacher, students
- We will negotiate scores per team member (team of 4 can get 34 points, distributed 9 + 9 + 8 + 8, or any combination determined by Scrum Master)
- If all team members receive the same score, laboratory coordinator will decide who presents the solution next week
- There is no upper limit to the points received
- THERE exists a lower limit for the points received: 50% of the sum of labs points
- Important: Make connections with the Java and Web technologies laboratories!!!

Laboratories (2)

- SCRUM
- We will use: Github, *BitBucket…*
- Trello, Google Drive

- Why to come to the labs? To be a part of a team, to earn points, learn to use project management techniques, to pass

Project

- Team work!
- It will involve:
  - Documentation, Modeling using UML diagrams
  - Implementation (main module, interface)
  - Testing, Evaluation, Documentation, Etc.
- Topics
  - Proposed by IT companies, PhD Students
  - CLEF 2024 (https://clef2024.imag.fr/): ImageCLEF, Plagiarism, QA, Social Book Search
- Frameworks and languages: C++, **Java**, Perl, Python, Java Script, C#

Project – Contest

- The best projects (7) will participate, during the last weeks of the semester (13-16), at a session of presentations
- At these sessions we will have a panel of professors and company representatives
- Students will receive bonuses, diplomas and prizes
- Learn & Earn – Amazon

Exam

- Without documentation
- 30 minutes

- Questions: grille + requiring answers on 2-3 lines + diagrams/schemes

- The emphasis will be on understanding the theoretical concepts presented in courses and practical ones used during the laboratories

Marks

- **Lab grade** – obtained during the first 6 labs (weekly assignments, teamwork)
- **Project grade** – obtained during the last 7 laboratories (a project where each member will contribute)
- **Exam grade** – 30 minutes, the focus will be on understanding the concepts presented
- **Final grade** = 10 x (Lab_grade + Project_grade + Exam_grade)/ MAX_GRADES_WITHOUT_BONUSES

Communication protocol

- Upon request, the solutions at certain stages will be sent by e-mail

- 3 requests:
  - Subject: **[SE] – Solution stage X**
  - Attachment: **TGZ , RAR, archive etc.** (**not ZIP**)
  - Content: **Signature Ionescu George, Year II, Group 2 E**

- **FAILURE to do so will result in downgrading (1 point for each missed requirement)**

Bibliography

- Course page Adrian Iftene http://profs.info.uaic.ro/~adiftene/Scoala/2024/SE/
- Ovidiu Gheorghieș page (he works with Adriana G.) http://profs.info.uaic.ro/~ogh/ip/
- Ian Sommerville: Software Engineering, Addison Wesley, 2001
- Craig Larman: Applying UML and Patterns, Addisson Wesley, 2002
- Erich Gamma, Richard Helm, Ralph Johnson, John Vissides: Design Patterns, Elements of Reusable Object-Oriented Software, Addisson Wesley, 1998
- Internet

Final results PE 2009-2022

- The continuation of these ideas and completing your graduation thesis in one of these areas
- Research articles and participation in conferences
- Long-term research bachelor, master, doctoral, post-doctoral (we have collaborations with universities in Spain, Italy, England, Ireland, France, etc.)

Results PE 2009 (1)

Evaluating Systems for Multilingual and Multimodal Information Access - Mozilla Firefox

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http://www.springer.com/computer+science/ai/book/978-3-642-04446-5

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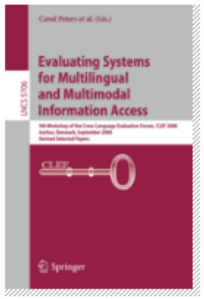

Evaluating Systems for Multilingual and Multimodal Information Access

9th Workshop of the Cross-Language Evaluation Forum, CLEF 2008, Aarhus, Denmark, September 17-19, 2008, Revised Selected Papers
Series: » Lecture Notes in Computer Science
Subseries: » Information Systems and Applications, incl. Internet/Web, and HCI, Vol. 5706

Peters, C.; Deselaers, Th.; Ferro, N.; Gonzalo, J.; Jones, G.J.F.; Kurimo, M.; Mandl, Th.; Peñas, A.; Petras, V. (Eds.)

2009, XXIV, 1002 p., Softcover
ISBN: 978-3-642-04446-5

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Results RE 2009 (2) – Corfu, Greece

Results PE 2009 (3) CLEF – WN

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http://www.clef-campaign.org/2009/working_notes/CLEF2009WN-Contents.html

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- Grid@CLEF
- MorphoChallenge
- Appendices

Español

Users' Image Seeking Behavior in a Multilingual Tag Environment
Miguel E Ruiz and Pok Chin

UAIC at iCLEF 2009: Analysis of Logs of Multilingual Image Searches in Flickr
Florin-Tudor Cristea, Vlad Alexa and Adrian Iftene

UNED at iCLEF 2009: Analysis of Multilingual Image Search Sessions
Víctor Peinado, Fernando López-Ostenero and Julio Gonzalo

Lexical Ambiguity in Cross-language Image Retrieval: a Preliminary Analysis
Borja Navarro-Colorado, Marcel Puchol-Blasco, Rafael M. Terol, Sonia Vázquez and Elena Lloret.

Users' Perceptions of Searching in FlickLing
Evgenia Vassilakaki, Frances Johnson, R.J. Hartley and David Randall

Multiple Language Question Answering (QA@CLEF)

Overview of ResPubliQA 2009: Question Answering Evaluation over European Legislation
Anselmo Peñas, Pamela Forner, Richard Sutcliffe, Álvaro Rodrigo, Corina Forăscu, Iñaki Alegria, Danilo Giampiccolo, Nicolas
Petya Osenova

CLEF is an activity of the TrebleCLEF Coordination Action

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Results PE 2009–2010

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COMPUTER SCIENCE

MULTILINGUAL INFORMATION ACCESS EVALUATION II. MULTIMEDIA EXPERIMENTS
Lecture Notes in Computer Science, 2010, Volume 6242/2010, 277-282, DOI: 10.1007/978-3-642-15751-6_34

Using SIFT Method for Global Topological Localization for Indoor Environments

Emanuela Boroş, George Roşca and Adrian Iftene

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Abstract

The paper represents a brief description of our system as one of the solutions to the problem of global topological localization for indoor environments. The experiment involves analyzing images acquired with a perspective camera mounted on a robot platform and applying a feature-based method (SIFT) and two main systems in order to search and classify the given images. To obtain acceptable results and improved performance improvement, the algorithm acquires two main maturity levels: one

Results ASET 2009

- Many themes continued as dissertation topics
- Contests, publications

Sesiunea Nationala **ITS 2010**
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Rezultate Finale

Nr.	Titlu Lucrare	Autor	Institutie
Premiul 1	Sisteme de creştere a siguranţei transportului rutier	Mircea CÎRCEAG, Isabela MOCANU, Răzvan MUNTEANU	an III, Facultatea Transporturi, Universitatea „Politehnica" din Bucureşti
Premiul 2	Sistem distribuit de Management al Traficului	Emilian NECULA, Raluca NECULA	Master, Facultatea de Informatica, Universitatea „A.I.Cuza" Iaşi
Premiul 2	Sisteme de informare a călătorilor	Camelia BUNEA, Andrei ŞORIGA	an IV, Facultatea Transporturi, Universitatea „Politehnica" din Bucureşti
Premiul 3	Sistem de alarmă cu transmiterea mesajului pe telefonul mobil	Alexandru STOICA	an III, Facultatea Transporturi, Universitatea „Politehnica" din Bucureşti
Premiul 3	Autovehiculul hibrid	Radu DINU, Ionuţ FLOREA, Diana TUDORACHE	an II, Facultatea Transporturi, Universitatea „Politehnica" din Bucureşti
Mentiune	Aplicatie GPS pentru PC	Andrei GEORGESCU	Master, Facultatea de Informatica, Universitatea „A.I.Cuza" Iaşi

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Using a genetic algorithm for optimizing the similarity aggregation step in the process of ontology alignment

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Faculty of Computer Science, "Al. I. Cuza" University, Iasi, Romania

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ABSTRACT

This paper addresses the increasingly encountered challenge of ontology alignment. Starting with basic similarity measures such as the syntactic similarity, represented by the Levenshtein or Jaro Distance, semantic similarities, which make use of WordNet and taxonomy similarities, our new system uses a genetic algorithm specially

2011-2012

- http://dblp.uni-trier.de/pers/hd/g/G=icirc=nsca:Alexandru=Lucian

2013–2014

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[+] [–] **Alexandra Siriteanu**

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[–] **2010 – today**

2014

[c2] Ralf Bierig, Cristina Serban, Alexandra Siriteanu, Mihai Lupu, Allan Hanbury: **A System Framework for Concept- and Credibility-Based Multimedia Retrieval.** ICMR 2014: 543

2013

[c1] Cristina Serban, Alexandra Siriteanu, Claudia Gheorghiu, Adrian Iftene, Lenuta Alboaie, Mihaela Breaban: **Combining Image Retrieval, Metadata Processing and Naive Bayes Classification at Plant Identification 2013.** CLEF (Working Notes) 2013

2015 – 2016

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[c1] Alexandru Calfa, Dragos Silion, Andreea Cristina Bursuc, Cornel Paul Acatrinei, Razvan Iulian Lupu, Alexandru Eduard Cozma, Cristian Padurariu, Adrian Iftene: **Using Textual and Visual Processing in Scalable Concept Image Annotation Challenge.** CLEF (Working Notes) 2015

Using Textual and Visual Processing in Scalable Concept Image Annotation Challenge

Alexandru Calfa, Dragoş Silion, Andreea Cristina Bursuc, Cornel Paul Acatrinei, Răzvan Iulian Lupu, Alexandru Eduard Cozma, Cristian Pădurariu, Adrian Iftene

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Abstract. This paper describes UAIC[1]'s system built for participating in the Scalable Concept Image Annotation challenge 2015. We submitted runs both for Subtask 1 (Image Concept detection and localisation) and for Subtask 2 (Generation of Textual Descriptions of Images). For the first subtask we created an ontology with relations between concepts and their synonyms, hyponyms and hypernyms and also with relations between concepts and related words. For the second subtask, we created a resource that contains triplets ($concept_1$, *verb*, $concept_2$), where *concepts* are from the list of concepts provided by the organizers and *verb* is a relation between concepts. With this resource we build sentences in which $concept_1$ is subject, *verb* is predicate and $concept_2$ is complement.

Using Machine Learning Techniques, Textual and Visual Processing in Scalable Concept Image Annotation Challenge

Alexandru-Gabriel Cristea, Mădălin-Marian Savoaia, Monica-Andreea Martac, Ionela Cristina Pătraş, Alexandru-Ovidiu Scutaru, Constantin-Emilian Covrig, Adrian Iftene

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Abstract. This paper describes UAIC[1]'s system built for participating in the Scalable Concept Image Annotation challenge 2016. We submitted runs for Subtask 1 (Image annotation and localisation), for Subtask 2 (Natural language caption generation) and for Subtask 3 (Content Selection). For the first subtask we used an ontology created last year with relations between concepts and their